

RAJEEV KULKARNI

Principal Decision Scientist | Life Sciences & Pharma Analytics | Patient & RWD Analytics | Payer/Provider Insights | Agentic AI
+91 7506115619 | kulkarni.rajeev@live.com | [LinkedIn](#)

PROFILE SUMMARY

Decision Science leader with 20+ years of experience, including 10+ years in healthcare analytics, 3 years dedicated to life sciences and pharma. Deep expertise in patient analytics (risk of readmission, mortality, etc.), oncology trials analytics (patient time to next treatment & trial criteria optimisation), and marketing analytics (MMM and market share analysis). Experienced in building and leading cross-functional teams across concurrent engagements with full delivery accountability and stakeholder advisory.

CORE COMPETENCIES

Clinical Trial Analytics · Generative AI & Multi-Agent Systems · Payer/Provider Analytics · Marketing Mix Modelling & Attribution · AI/ML Roadmap & Strategy · Applied ML & Statistical Modelling · Model Governance & Explainability · Thought Leadership & Publications · Cross-Functional Team Leadership · Stakeholder Advisory & Executive Communication

TECHNICAL SKILLS

Languages & Tools: Python | R | SQL | AWS SageMaker

AI & Gen AI: Prompt & Context engineering | Multi-Agent Gen AI Systems

Analytics: Risk Stratification | Survival Models | Marketing Mix Models | Model Governance & Explainability

WORK EXPERIENCE

ConcertAI | Principal Data Scientist | 2023 – Present

- Led the AI/ML programme for a real-world data platform serving life sciences sponsors conducting oncology clinical trials — working with claims, EHR, physician-curated disease status notes, labs, conditions, and genomics data.
- Rescued a stalled 1+ year programme projected at a 2+ year timeline — proposed an alternative approach to model building and deployed SME-validated models across 13 cancer indications in 8 months with a team of 5.
- Models were time-to-next-treatment (TTNT) models achieving C-index ≥ 0.8 — integrated into a sponsor-facing product enabling early identification of patients eligible for clinical trial enrolment, deployed across 40+ Phase III trial sites.
- Program generated three ASCO publications, with the 2025 publication accepted as a poster.
- Productionised a multi-agent GenAI system automating clinical trial eligibility optimisation — replacing a time-intensive manual SME process, achieving SME-AI recommendation concordance F1 > 0.9 across 30 trials; submitted for ASCO 2026.

Integra Connect | Sr. Data Scientist | 2017 – 2020

- Built predictive models for mortality, readmission, and skilled nursing facility utilisation under the Oncology Care Model (OCM) — working with claims and EHR data and deployed across 1,500+ oncology practices participating in the OCM programme.
- Models enabled early identification of patients at risk of expensive interventions, allowing practices to intervene earlier and remain under agreed payer price thresholds — the core economic mechanism of the OCM programme.
- Analytics contributions were foundational to the OCM programme generating \$250M in revenue and \$100M in cost savings across participating provider practices.

Philips | Senior Manager, Marketing Data Science | 2020 – 2023

- Led cross-market Marketing Mix Modelling (MMM) delivery across US and EU markets — informing multi-million-dollar budget allocation decisions through model-driven insights.
- Developed brand performance models supporting market share growth across business units.

CitiusTech | Manager, Data Science | 2014 – 2017

- Earned internal referrals through early delivery of complex analytics - growing a single payer engagement from \$80K to \$1M.
- Led proposals, staffing strategy, mentoring, and programme governance across client engagements.

Mercury Data Systems | Systems Engineer | 2006 – 2013

- Built multi-sensor systems for navigation in GPS-denied environments.

SELECT PUBLICATIONS (ASCO) — THOUGHT LEADERSHIP IN LIFE SCIENCES AI

- A Multi-Agent AI System Integrating Real-World Data and Competitive Intelligence for Automated Optimisation of Trial Eligibility Criteria — ASCO 2026 (submitted)
- Machine Learning model to forecast patient availability for oncology clinical trials — ASCO 2025 (Poster)
- Dynamic ML model to forecast patient availability for clinical trials — ASCO 2024
- AI-based vs EMR-based clinical trial eligibility screening — ASCO 2024

EDUCATION

MSEE (Electrical Engineering) · Michigan Technological University, USA

BE (Electronics & Telecommunications) · Mumbai University, India

Details of Healthcare Projects (contd.)

PROJECT DETAILS

LIFE SCIENCES | ConcertAI

Patient Matching To Oncology Trials (Published ASCO 2025 Poster)

Automating **eligibility criteria** assessment for oncology clinical trials is an emerging application of machine learning (ML). However, machine learning applications to predict **patient availability** – the likelihood of a patient beginning a new treatment (time to next treatment) in a prespecified time window – are not well described. We used a large clinicogenomic database of patients diagnosed with solid cancer indications to train ML model to predict patient availability for clinical trials.

METHODOLOGY

- Retrospective study using ConcertAI Oncology Research database, enriched with unstructured data; cohort included solid cancer patients without second malignancies, followed from metastasis diagnosis to last activity/death.
- Line of therapy derived from structured medications via expert rules.
- To create labels, a random observation date was set within follow-up to label patients as “event” (new treatment after the date) or censored (no new treatment after the date).
- Time to event (TTE) measured from random observation to new treatment start (TTNT for events) or end date (censored).
- Generated over 2000 features from biomarkers (e.g., PTEN, KRAS), ECOG, staging, medications, disease status, and imaging to build Time to Next Treatment (TTNT) models.
- Temporal validation of the models was performed by setting up a simulated index date and predicting the probability of patient beginning a new treatment within 60 days of the simulated index date.
- Patients receiving new treatment within the 60 days were true positives.

RESULTS

- Models were built on 90K patients across 12 solid cancer indications with 54% patients starting new treatment.
- Model with C-index = 0.73, weighted F1 score of 67%.

When combined with a liquid cancer TTNT model, the programme achieves C-index>0.8 across 13 indications.

Multi-Agent AI System Integrating Real-World Data and Competitive Intelligence for Automated Optimisation of Trial Eligibility Criteria (submitted for ASCO 2026)

Designing eligibility criteria is central to clinical trial feasibility but requires laborious manual review across multiple dimensions, including competitive intelligence (CI), standard-of-care real-world data (SoC-RWD), and evolving therapeutic trends (ET). We developed a multi-agent AI framework to automate this process and generate evidence-based recommendations for optimizing trial design.

METHODOLOGY:

- Built the SoC-RWD and Emerging Therapeutic Trends agents.
- Agents were provided draft inclusion / exclusion criteria for the trial.
- Internal APIs and AI workflows extract line of treatment (LOT) for patients eligible for the draft trial criteria from the ConcertAI Oncology Research database.
- SoC-RWD agent benchmarks the draft criteria with the LOT data for treatment utilisation.
- Emerging Trends agent benchmarks the draft criteria with *changes* in the LOT data to identify evolution of treatment utilisation.
- Each agent issues criteria level recommendations with rationale that are consolidated into a unified output.
- Agent recommendations were validated by two clinical scientist independently, rating each recommendation on a scale of 1-5.

RESULTS:

- Agents recommendations were validated over 30 oncology trials (NSCLC, breast and colorectal indications).
- Agent clinical scientist concordance (scientist agreement with agent recommendation) was > 95%.
- The SoC and ET agents achieved weighted F1 > 0.9 across these trials.
- Agents produced deterministic results – for same set of criteria and real world data, agents produce the *same* output 9/10 trials and *correct* outputs 10/10 trials.

PROVIDER ONCOLOGY | IntegraConnect

Oncology treatments under value based care (VBC) insurance instruments are capped for each 6 month episode of patient treatment to ensure high quality of care and affordable care (at the population level) under the Oncology Care Model (OCM). IntegraConnect's oncology EHR was designed to support the reporting requirements imposed by the OCM. Predictive models were added to the oncology EHR to enhance provider's ability to stratify patients based on their risk of expensive interventions such as readmissions, mortality and continued care under skilled nursing facility (SNF).

METHODOLOGY:

- Claims data was used to identify patient cohort and define episodes of care.
- Patient history was extracted from EHR – conditions, treatments, adverse events, toxicities, lab values and vitals – for the cohort.
- Labels were extracted from claims data (readmissions and SNF), or demographic data (mortality).
- Patient history was featurised to build classification models for the three outcomes.

RESULTS:

- Model AUC > 0.9 and weighted F1 scores > 0.8 for the models.
- Models integrated into provider workflows across 1,500+ oncology providers.
- OCM programme generated \$250M in revenue and \$100M in cost savings.